

COMSATS University Islamabad

Attock Campus

Department of Computer Science

Library Management System

Database Design & Implementation

Course Database Systems (Lab)	Instructor Mr. Shahzad Rizwan
Database LibraryDB (MySQL)	Academic Year 2024 – 2025

1. Project Overview

The Library Management System (LMS) is a fully-featured relational database application developed for the COMSATS University Islamabad, Attock Campus as part of the Database Systems Lab course. It provides a structured, normalized, and performant backend for managing all core library operations — from book cataloguing and member registration to loan tracking, fine management, and book requests.

1.1 Objectives

- Design a normalized relational database schema for a real-world library.
- Implement multi-valued attributes via dedicated associative tables.
- Track books, members, loans, fines, inventory, and staff.
- Write analytical SQL queries covering joins, aggregations, subqueries, and window functions.
- Build a responsive web-based frontend dashboard for visualization.

1.2 Technology Stack

Layer	Technology	Purpose
Database Engine	MySQL 8.x	Relational storage & SQL processing
Schema	DDL (library_schema.sql)	Tables, constraints, indexes
Data	DML (library_data.sql)	Sample data population
Queries	SQL (library_queries.sql)	Reporting & analytics — 22 queries
Frontend	HTML / CSS / Vanilla JS	Dashboard UI (library_frontend.html)

2. Database Schema

The database LibraryDB consists of 16 tables organized into 8 functional modules. All multi-valued attributes (authors, categories, contacts, addresses, sections) are stored in separate tables to achieve First Normal Form (1NF) compliance and beyond.

2.1 Entity-Relationship Overview

The diagram below describes the high-level relationships between all major entities in the system:

Entity-Relationship Map — LibraryDB			
BOOKS		MEMBERS	STAFF
Books ← Book_Authors Books ← Book_Categories Books ← Book_Publishers		Members ← Member_Contacts Members ← Member_Address	Staff ← Staff_Contacts Staff ← Staff_Address Staff ← Staff_Sections
TRANSACTIONS		INVENTORY	
Loans (Members ↔ Books) Fines (Members → Books) Requests (Members → Books)		Inventory (Books → Stock) Publishers (via Book_Publishers)	

2.2 Table Schemas

Below are the detailed column-level schemas for all 16 tables in LibraryDB.

Books

Column Name	Data Type	Constraints	Description
Book_ID	INT	PRIMARY KEY, AUTO_INCREMENT	Unique book identifier
Title	VARCHAR (255)	NOT NULL	Title of the book
Edition	VARCHAR (50)	NULLABLE	Edition number (e.g., 3rd, 7th)

Column Name	Data Type	Constraints	Description
ISBN	VARCHAR (20)	UNIQUE, NOT NULL	International Standard Book Number

Book_Authors

Column Name	Data Type	Constraints	Description
Book_ID	INT	PK, FK → Books	Reference to parent book
Author	VARCHAR (150)	PK, NOT NULL	Author name (composite PK allows multiple)

Book_Categories

Column Name	Data Type	Constraints	Description
Book_ID	INT	PK, FK → Books	Reference to parent book
Category	VARCHAR (100)	PK, NOT NULL	Category tag (e.g., Computer Science, AI)

Publishers

Column Name	Data Type	Constraints	Description
Publisher_ID	INT	PRIMARY KEY, AUTO_INCREMENT	Unique publisher ID
Name	VARCHAR (150)	NOT NULL	Publisher name
Address	VARCHAR (255)	NULLABLE	Office address
Contact_Number	VARCHAR (20)	NULLABLE	Contact phone number

Book_Publishers

Column Name	Data Type	Constraints	Description
Book_ID	INT	PK, FK → Books	Book reference
Publisher_ID	INT	PK, FK → Publishers	Publisher reference (junction table)

Members

Column Name	Data Type	Constraints	Description
Member_ID	INT	PRIMARY KEY, AUTO_INCREMENT	Unique member ID
Name	VARCHAR (150)	NOT NULL	Full name of member
Membership_Type	ENUM	NOT NULL	Student / Faculty / Staff / Guest

Member_Contacts

Column Name	Data Type	Constraints	Description
Member_ID	INT	PK, FK → Members	Parent member reference
Phone_Number	VARCHAR (20)	PK, NOT NULL	Contact phone (supports multiple)

Member_Address

Column Name	Data Type	Constraints	Description
Member_ID	INT	PK, FK → Members	Parent member reference (1-to-1)
Address	VARCHAR (255)	NOT NULL	Residential address

Staff

Column Name	Data Type	Constraints	Description
Staff_ID	INT	PRIMARY KEY, AUTO_INCREMENT	Unique staff ID
Name	VARCHAR (150)	NOT NULL	Full name

Column Name	Data Type	Constraints	Description
Role	<code>VARCHAR(100)</code>	NOT NULL	Job role (Librarian, Assistant, etc.)

Staff_Sections

Column Name	Data Type	Constraints	Description
Staff_ID	<code>INT</code>	PK, FK → Staff	Staff reference
Assigned_Section	<code>VARCHAR(100)</code>	PK, NOT NULL	Library section assigned to staff

Inventory

Column Name	Data Type	Constraints	Description
Inventory_ID	<code>INT</code>	PRIMARY KEY, AUTO_INCREMENT	Unique inventory record ID
Book_ID	<code>INT</code>	UNIQUE, FK → Books	One-to-one with Books
Title	<code>VARCHAR(255)</code>	NOT NULL	Denormalized title for quick display
Shelf_Location	<code>VARCHAR(50)</code>	NULLABLE	Physical shelf code (e.g., A-1-10)
Stock_Count	<code>INT</code>	NOT NULL, CHECK ≥ 0	Currently available copies
Total_Count	<code>INT</code>	NOT NULL, CHECK ≥ 0	Total copies ever acquired

Loans

Column Name	Data Type	Constraints	Description
Loan_ID	INT	PRIMARY KEY, AUTO_INCREMENT	Unique loan record
Member_ID	INT	FK → Members (RESTRICT)	Borrowing member
Issued_Book_ID	INT	FK → Books (RESTRICT)	Book being loaned
Issue_Date	DATE	NOT NULL	Date book was issued
Due_Date	DATE	NOT NULL	Date book must be returned
Return_Date	DATE	DEFAULT NULL	Actual return date (NULL = active loan)

Fines

Column Name	Data Type	Constraints	Description
Fine_ID	INT	PRIMARY KEY, AUTO_INCREMENT	Unique fine record
Member_ID	INT	FK → Members (RESTRICT)	Member who incurred the fine
Overdue_Book_ID	INT	FK → Books (RESTRICT)	Book that was overdue
Overdue_Days	INT	NOT NULL, CHECK > 0	Number of days overdue
Fine_Amount	DECIMAL(8, 2)	NOT NULL, CHECK ≥ 0	Monetary fine in currency units
Payment_Status	ENUM	NOT NULL	Paid / Unpaid

Requests

Column Name	Data Type	Constraints	Description
Request_ID	INT	PRIMARY KEY, AUTO_INCREMENT	Unique request record
Member_ID	INT	FK → Members (RESTRICT)	Requesting member
Requested_Book_ID	INT	FK → Books (RESTRICT)	Book being requested
Request_Date	DATE	NOT NULL, DEFAULT CURRENT_DATE	Date of request
Request_Status	ENUM	NOT NULL	Pending / Approved / Rejected / Fulfilled

3. Normalization Analysis

The LibraryDB schema adheres to the Third Normal Form (3NF) throughout. The following table summarizes the normalization decisions made for each module.

Entity	1NF	2NF/3NF	Design Decision
Books	✓ Multi-valued attributes removed	✓ No partial/transitive deps	Authors & Categories in separate tables
Members	✓ Contacts & Address extracted	✓ No transitive dependencies	Member_Contacts & Member_Address tables
Staff	✓ Contacts, Address, Sections split	✓ All non-key attributes depend on PK	3 satellite tables for Staff entity
Loans	✓ Atomic columns only	✓ All facts depend on Loan_ID	Return_Date = NULL for active loans
Fines	✓ One fine per row	✓ Fine_Amount depends on Fine_ID	Payment_Status as ENUM, not a flag
Inventory	✓ Atomic stock counts	✓ Stock_Count & Total_Count are facts	UNIQUE constraint on Book_ID (1:1)
Requests	✓ One request per row	✓ Status depends on Request_ID	Request_Status ENUM for state machine
Publishers	✓ Atomic address & contact	✓ No transitive deps	Book_Publishers junction table for M:N

4. Sample Data

4.1 Books (10 records)

ID	Title	Ed.	Authors	Publisher
101	Data Structures	3rd	Ellis Horowitz, Sartaj Sahni	Pearson
102	Database Systems	7th	C.J. Date, A. Kannan	McGraw Hill
103	Algorithms	5th	Thomas H. Cormen, Charles E. Leiserson	MIT Press
104	Operating Systems	4th	Abraham Silberschatz	Prentice Hall
105	Computer Networks	6th	Andrew Tanenbaum	Prentice Hall
106	Artificial Intelligence	2nd	Stuart Russell, Peter Norvig	MIT Press
107	Software Engineering	9th	Ian Sommerville	Wiley
108	Discrete Mathematics	1st	Kenneth H. Rosen	Pearson
109	Linear Algebra	3rd	Gilbert Strang	MIT Press
110	Calculus	8th	James Stewart	Wiley

4.2 Members (8 records)

ID	Name	Type	Phone	Address
1	John Doe	Student	555-1234, 555-5678	123 Elm Street
2	Jane Smith	Faculty	555-9876	456 Maple Avenue
3	Ali Hassan	Student	555-1111	789 Oak Lane

ID	Name	Type	Phone	Address
4	Sara Ahmed	Student	555-2222	12 Pine Road
5	Dr. Malik	Faculty	555-3333, 555-4444	34 Cedar Blvd
6	Usman Tariq	Staff	555-5555	56 Walnut Drive
7	Ayesha Khan	Student	555-6666	78 Birch Court
8	Bilal Chaudhry	Student	555-7777	90 Spruce Terrace

4.3 Active Loans

ID	Member	Book	Issued	Due	Status
1	John Doe	Data Structures	2024-12-01	2024-12-15	Active
2	John Doe	Database Systems	2024-12-03	2024-12-17	Active
3	Jane Smith	Algorithms	2024-12-05	2024-12-20	Active
5	Sara Ahmed	Computer Networks	2024-12-10	2024-12-24	Active
7	Ayesha Khan	Discrete Mathematics	2024-12-07	2024-12-21	Active
8	Bilal Chaudhry	Linear Algebra	2024-12-08	2024-12-22	Active

4.4 Fines Summary

Member	Book	Days	Amount	Status
John Doe	Data Structures	5	\$10.00	Unpaid
John Doe	Database Systems	3	\$10.00	Unpaid

Member	Book	Days	Amount	Status
Jane Smith	Algorithms	7	\$7.00	Paid
Sara Ahmed	Computer Networks	2	\$4.00	Unpaid
Bilal Chaudhry	Linear Algebra	4	\$8.00	Unpaid

5. SQL Queries Reference

The project contains 22 SQL queries organized into 8 sections. Each query is documented below with its purpose and key SQL techniques used.

Q#	Section	Purpose	Key Technique
Q1	Books	List all books with authors and categories	GROUP_CONCAT, Multi-JOIN
Q2	Books	Find all books by a specific author	JOIN + LIKE filter
Q3	Books	Find all books in a specific category	JOIN + WHERE
Q4	Books	Books with publisher info	3-table JOIN
Q5	Members	All members with contact and address	LEFT JOIN, GROUP_CONCAT
Q6	Members	Members currently holding loans	DISTINCT + NULL check
Q7	Members	Members with unpaid fines	SUM aggregate + filter
Q8	Loans	All active loans with overdue days	DATEDIFF(CURDATE(), ...)
Q9	Loans	Overdue loans past due date	Date comparison + ORDER BY
Q10	Loans	Loan history for a specific member	CASE expression
Q11	Loans	Total loans per member (ranking)	COUNT + LEFT JOIN + ORDER
Q12	Inventory	Inventory status with availability %	Computed column (ROUND %)
Q13	Inventory	Books with low stock (< 3 copies)	WHERE threshold filter
Q14	Fines	All fines with member and book details	3-table JOIN

Q#	Section	Purpose	Key Technique
Q15	Fines	Total fines collected vs outstanding	GROUP BY Payment_Status
Q16	Requests	All pending requests	WHERE + JOIN + ORDER
Q17	Requests	Most requested books	COUNT + GROUP BY + ORDER
Q18	Staff	All staff with their sections	4-table JOIN + GROUP_CONCAT
Q19	Advanced	Most borrowed books (Top 5)	COUNT + LIMIT
Q20	Advanced	Books never borrowed	LEFT JOIN + IS NULL
Q21	Advanced	Summary dashboard stats	6 scalar subqueries
Q22	Advanced	Members who borrowed more than average	Correlated subquery + HAVING

5.1 Highlighted Query — Dashboard Stats (Q21)

This query uses six scalar subqueries to produce a single-row summary suitable for a dashboard:

```
SELECT
    (SELECT COUNT(*) FROM Books) AS Total_Books,
    (SELECT COUNT(*) FROM Members) AS Total_Members,
    (SELECT COUNT(*) FROM Loans WHERE Return_Date IS NULL) AS Active_Loans,
    (SELECT COUNT(*) FROM Fines WHERE Payment_Status = 'Unpaid') AS
    Unpaid_Fines,
    (SELECT SUM(Fine_Amount) FROM Fines WHERE Payment_Status = 'Unpaid') AS
    Outstanding,
    (SELECT COUNT(*) FROM Requests WHERE Request_Status = 'Pending') AS
    Pending_Requests;
```

5.2 Highlighted Query — Above-Average Borrowers (Q22)

This advanced query uses a correlated subquery with HAVING to identify power users:

```
SELECT m.Name, m.Membership_Type, COUNT(l.Loan_ID) AS Total_Loans
FROM Members m
JOIN Loans l ON m.Member_ID = l.Member_ID
GROUP BY m.Member_ID, m.Name, m.Membership_Type
HAVING COUNT(l.Loan_ID) > (
    SELECT AVG(loan_count) FROM (
        SELECT COUNT(*) AS loan_count FROM Loans GROUP BY Member_ID
    ) sub
);
```


6. Inventory & Statistics

6.1 Inventory Status

Book Title	Shelf	Stock	Total	Availability
Data Structures	A-1-10	5	10	50.0%
Database Systems	B-2-20	3	8	37.5%
Algorithms	A-2-05	6	12	50.0%
Operating Systems	C-1-15	4	7	57.1%
Computer Networks	C-2-10	2	5	40.0%
Artificial Intelligence	D-1-08	7	10	70.0%
Software Engineering	D-2-12	3	6	50.0%
Discrete Mathematics	E-1-03	8	15	53.3%
Linear Algebra	E-2-07	5	9	55.6%
Calculus	F-1-20	4	11	36.4%

6.2 Dashboard KPIs

10	8	6	4	\$32.00	4
Total Books	Total Members	Active Loans	Unpaid Fines	Outstanding	Pending Requests

7. Frontend Dashboard

The project includes a single-page web dashboard (library_frontend.html) built with vanilla HTML, CSS, and JavaScript. It provides a visual interface for exploring all database entities and query results.

7.1 Features

- Dark-themed responsive dashboard with sidebar navigation
- Dedicated sections: Books, Members, Staff, Loans, Fines, Inventory, Requests
- KPI cards showing real-time summary statistics
- Color-coded status indicators (Active/Returned loans, Paid/Unpaid fines)
- Availability percentage bars in inventory view
- Mobile-friendly layout with collapsible sidebar
- DM Sans / DM Serif Display / DM Mono Google Fonts typography

7.2 UI Structure

Section	Data Displayed	Key UI Elements
Dashboard	KPI cards (Books, Members, Loans, Fines)	Stat cards, color-coded badges
Books	All 10 books with authors, edition, ISBN	Table with multi-author display
Members	Member list with type, phone, address	Membership type color badge
Staff	Staff roles, sections, contacts	Section tags, role labels
Loans	Active & returned loans with due dates	Overdue highlighting in red
Fines	Fine records with payment status	Paid (green) / Unpaid (red) badges
Inventory	Stock counts and availability %	Progress-style percentage display
Requests	Book requests with status	Pending / Approved / Fulfilled badges

8. Indexes & Performance

The schema includes 6 optimized indexes on frequently queried foreign key and filter columns to ensure fast query execution:

Index Name	Column	Purpose
idx_books_title	Books (Title)	Speeds up title-based searches
idx_loans_member	Loans (Member_ID)	Faster member loan history lookups
idx_loans_book	Loans (Issued_Book_ID)	Faster book borrow history
idx_fines_member	Fines (Member_ID)	Quick fine retrieval per member
idx_requests_member	Requests (Member_ID)	Request lookup by member
idx_requests_status	Requests (Request_Status)	Filtering pending/approved requests

8.1 Constraint Summary

Constraint Type	Count	Example	Enforcement
PRIMARY KEY	16 (one per table)	Books(Book_ID)	Uniqueness + clustered index
FOREIGN KEY	13 references	Loans(Member_ID) → Members	Referential integrity (CASCADE/RESTRICT)
UNIQUE	2 columns	Books(ISBN), Inventory(Book_ID)	No duplicate ISBNs or duplicate inv. entries
NOT NULL	30+ columns	Books(Title), Members(Name)	Data completeness guarantee
CHECK	3 constraints	Stock_Count ≥ 0, Overdue_Days > 0	Business rule enforcement

Constraint Type	Count	Example	Enforcement
ENUM	3 columns	Membership_Type, Payment_Status, Request_Status	Valid state enforcement
DEFAULT	3 columns	Return_Date = NULL, Request_Date = CURRENT_DATE	Sensible defaults

9. Setup & Installation Guide

9.1 Prerequisites

- MySQL 8.0 or higher installed
- MySQL Workbench, DBeaver, or any MySQL client
- A modern web browser (Chrome, Firefox, Edge) for the frontend

9.2 Step-by-Step Setup

Step	Action	Command / Notes
1	Clone / Download	Download all 4 SQL files and the HTML file from the repository.
2	Run Schema	Execute library_schema.sql — this creates LibraryDB and all 16 tables.
3	Load Data	Execute library_data.sql — this populates all tables with sample records.
4	Run Queries	Execute library_queries.sql — review all 22 analytical queries.
5	Open Frontend	Open library_frontend.html in any browser — no server required.

9.3 File Structure

File	Description
library_schema.sql	DDL — Creates database, all 16 tables, indexes, and constraints
library_data.sql	DML — Inserts sample data for all modules (books, members, staff, etc.)
library_queries.sql	22 SQL queries covering all functional areas + advanced analytics
library_frontend.html	Self-contained single-page dashboard — open directly in browser

File	Description
README.md	Quick-start guide and project overview for GitHub

10. Conclusion

The Library Management System demonstrates a complete, production-quality relational database design from schema definition through data population, analytical querying, and frontend visualization.

Key achievements of this project include:

- Full 3NF normalized schema with 16 tables handling all library operations
- Proper use of CASCADE and RESTRICT foreign key actions to maintain data integrity
- 22 diverse SQL queries covering simple selects, multi-table joins, GROUP BY aggregations, correlated subqueries, and date arithmetic
- A responsive, dark-themed frontend dashboard without any external framework dependencies
- Comprehensive indexing strategy targeting frequently filtered columns

This project was completed as part of the Database Systems Lab course at COMSATS University Islamabad, Attock Campus, under the guidance of Instructor Mr. Shahzad Rizwan.